- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

10 A ball is thrown upwards with a velocity of $29.4\,\mathrm{m\,s^{-1}}$.

(a) Show that the ball reaches its maximum height after 3 s. **[1]**

(b) Sketch a velocity-time graph for the first 5 s of motion. **[2]**

(c) Calculate the speed of the ball 5 s after it is thrown. **[3]**

A second ball is thrown at $u\,\mathrm{m\,s^{-1}}$ at an angle of $\alpha°$ above the horizontal. It reaches the same maximum height as the first ball.

(d) Use this information to write down

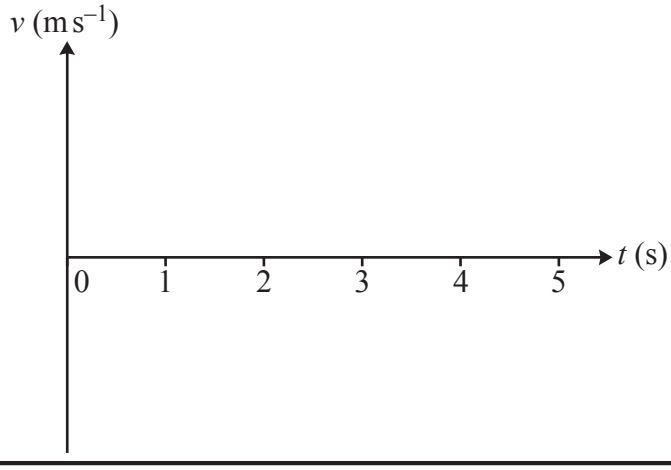
- the vertical component of the second ball's initial velocity,
- the time taken for the second ball to reach its greatest height. **[2]**

This second ball reaches its greatest height at a point which is 48 m horizontally from the point of projection.

(e) Calculate the values of u and α. **[3]**

10(a)

10(b)



10(c)

10(d)

Vertical component of the second ball's initial velocity =

Time taken for the second ball to reach its greatest height =

10(e)	
	$u =$
	$\alpha =$